

AI Research Master's & PhD Roadmap (For Md Jobayed Hossain Rabbi)

Goal: Complete current Master of IT (AI), build a strong AI portfolio, secure a Research Master's opportunity, and progress toward a funded PhD.

Phase 1: Python Foundation (1-2 Months)

- Variables
- Data Types
- If/Else
- Loops
- Functions
- Lists
- Dictionaries
- File Handling
- OOP Basics

Learning Sources:

<https://docs.python.org/3/tutorial/>

<https://www.youtube.com/watch?v=rfscVS0vtbw>

<https://www.hackerrank.com/domains/python>

Phase 2: Data Analysis

- NumPy
- Pandas
- Matplotlib
- Data Cleaning
- Data Visualization

Learning Sources:

<https://numpy.org/learn/>

https://pandas.pydata.org/docs/getting_started/

<https://matplotlib.org/stable/tutorials/index.html>

Phase 3: SQL for Data Science

- SELECT
- JOIN
- GROUP BY
- ORDER BY
- Aggregate Functions

Learning Sources:

<https://www.w3schools.com/sql/>

<https://sqlbolt.com/>

Phase 4: Machine Learning

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- Scikit-Learn

Learning Sources:

<https://scikit-learn.org/stable/tutorial/>

<https://www.khanacademy.org/math/statistics-probability>

Phase 5: AI Projects

- Student Performance Prediction
- House Price Prediction
- Loan Approval Prediction

Learning Sources:

<https://www.kaggle.com/>

Phase 6: Deep Learning

- Neural Networks
- CNN
- RNN
- TensorFlow
- PyTorch

Learning Sources:

<https://www.tensorflow.org/tutorials>

<https://pytorch.org/tutorials/>

Phase 7: Computer Vision

- Image Classification
- Object Detection

Learning Sources:

<https://www.kaggle.com/learn/computer-vision>

Phase 8: NLP

- Tokenization
- TF-IDF
- Sentiment Analysis
- Fake News Detection

Learning Sources:

<https://www.kaggle.com/learn/natural-language-processing>

Phase 9: Generative AI & LLM

- OpenAI API
- LangChain
- RAG
- Vector Databases

Learning Sources:

<https://platform.openai.com/docs>

<https://python.langchain.com/docs/>

Phase 10: Portfolio & Research

- 8+ GitHub Projects
- LinkedIn Posts
- Google Scholar Reading
- Professor Outreach

Learning Sources:

<https://scholar.google.com>

<https://arxiv.org>

Suggested Timeline

Sep-Dec: Python, Pandas, NumPy, SQL

Jan-Mar: Machine Learning + 3 Projects

Apr-Jun: NLP + Deep Learning

Jul-Dec: LLM Projects, Supervisor Contact, Research Master Applications

Potential CDU Researchers

Niusha Shafi Abady - <https://researchers.cdu.edu.au/en/persons/niusha-shafi-abady/>

CDU Research Directory - <https://researchers.cdu.edu.au/en/persons/>

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